Delta Sleep Inducing Peptide (DSIP)

100–500 mcg SC or intranasally near bedtime

Delta sleep-inducing peptide (DSIP) is a synthetic nine-amino-acid peptide based on material originally isolated from rabbit cerebral venous blood in slow-wave-sleep experiments. [1,2] It is investigated for sleep continuity. Small human IV studies have produced mixed sleep findings. [4–7,10,13,30] DSIP's endogenous identity and specific receptor remain unresolved. [19] Experimental findings implicate neuroendocrine signaling, inhibitory–excitatory balance, and cellular stress responses. [12,18,20]

💉💉 **Brief IV administration was generally well tolerated in small insomnia studies. Cardiovascular reactions in other clinical settings and the separate risks of repeated SC exposure require endpoint-specific assessment. [21,23]**

🅾️ **Sleep efficacy is inconclusive. Small human studies range from improved sleep measures to weak effects of limited therapeutic significance. [4–7,10,13,30]**

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

Deep sleep, also called slow-wave sleep or stage N3, is the deepest non-rapid eye movement (non-REM) stage and is difficult to interrupt. It is associated with restorative processes; the amount of slow-wave sleep is also associated with growth hormone secretion. [26,27] Rapid eye movement (REM) sleep features rapid eye movements, dreaming, and temporary inhibition of arm and leg muscle activity. Both REM and non-REM sleep contribute to normal sleep cycling and memory processes. Lighter non-REM stages precede deep sleep. [26]

Small human studies observed tolerability of slowly administered IV DSIP over brief treatment periods. [4–7,13,23] This finding is distinct from longer-term hepatic, renal, cardiovascular, immunological, reproductive, and proliferative outcomes.

Human evidence for sleep improvement is mixed. Several small early IV studies reported improvements in sleep duration, efficiency, or continuity. [4–6] The two 1981 insomnia reports describe the same patient cohort and should not be counted as independent replication. [3,5,23] The positive 1987 study used seven successive IV treatments and compared results with baseline and external healthy controls; it did not include a concurrent randomized placebo-treatment arm. [10,23] A separate 1987 double-blind crossover study found sleep changes of little clinical significance. [30] The 1992 double-blind study found weak objective changes and no subjective sleep-quality benefit. Overall efficacy remains inconclusive. [13]

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Dosing Evidence and Patterns

DSIP degrades rapidly. [11] Early human sleep studies used slow intravenous (IV) administration and reported that administration rate influenced the observed response. [6,8] Subcutaneous (SC) administration could plausibly slow absorption relative to IV administration.

Historical human experiments used slowly administered IV DSIP at 25 nmol/kg. [4–6,13,30] With a molecular weight of approximately 848.8 g/mol, this equals about 21.2 mcg/kg, or approximately 1.5 mg for a 70-kg person. [2,23]

An early review described U-shaped relationships between DSIP effects and dose. This pattern makes a simple assumption that more exposure produces more benefit unreliable. [8]

Human SC bioavailability and PK have not been established for DSIP. [23] As a speculative sensitivity model, assuming 30%–80% SC bioavailability would make approximately 1.9–5 mg SC arithmetically equivalent to 1.5 mg IV in total systemic exposure, if clearance were also unchanged. The 30%–80% range is an assumed input, not a measured DSIP value.

Published human studies examined short IV regimens including three, four, and seven consecutive treatment nights or afternoons. [6,10,13,30] A small open study administered ten injections to seven patients. [7]

Speculative Experimental Dosing

In spite of the larger IV doses used in studies, the typical practice found today is a much smaller dose, typically taken in intranasally before bedtime.

- **100–500 mcg SC or intranasally near bedtime**

Continuous use and cycling are both speculative experimental schedules discussed.

Human IV studies reported effects beginning after approximately one to two hours and, in some experiments, delayed effects on subsequent sleep. [4–6] A slower absorption profile after SC administration is plausible; a specific pre-sleep interval, onset time, or duration for SC or intranasal use remains speculative.

Benefits

Reported effects vary substantially across small human studies, from measurable sleep improvements to weak effects of limited clinical significance. [4–7,10,13,30] Differences in study design, population, administration route, dose, and timing could contribute to this variation; as could placebo effects in uncontrolled, unblinded studies. [8,23]

- **Sleep architecture:** Effects are inconclusive. Some human studies reported improved sleep efficiency, latency, or continuity, while controlled findings were weak in other studies. [4–

6,10,13,30] Increased delta activity in rats does not establish increased human slow-wave sleep; effects vary across experimental systems. [8,16,19]

- **Neuroendocrine modulation:** IV DSIP reduced plasma adrenocorticotropic hormone-like immunoreactivity (ACTH-LI) for at least three hours in a randomized crossover, without changing cortisol. Broader modulation of hypothalamic-pituitary-adrenal (HPA) function and stress-related arousal is plausible. [12]
- **Daytime function:** Small human studies reported improved daytime alertness or performance after IV DSIP, although the positive 1987 study lacked a concurrent placebo-treatment group. [6,10,23] Mood improvement was reported in the seven-patient open study. [7]

Potential Applications

- **Insomnia or fragmented sleep**
- **Stress-related sleep disruption**
- **Recovery support**
- **Non-benzodiazepine sleep research**

Cautions

- **Pregnancy and breastfeeding:** Reproductive, developmental, and lactation safety remain insufficiently characterized. Estrous-cycle observations in adult mice do not establish pregnancy or infant safety. [23,25]
- **Severe or unstable psychiatric illness:** Effects on psychiatric symptoms remain insufficiently characterized; changes in arousal and neuroendocrine signaling make extrapolation from small insomnia studies inappropriate. [12,21,23]
- **Concurrent sedative-hypnotics, central nervous system depressants, or other investigational sleep agents:** Altered sedation, arousal, or sleep architecture is plausible. The direction may differ by agent: DSIP lightened isoflurane anesthesia in a small randomized study, rather than uniformly increasing sedation. [21]
- **Severe hepatic or renal impairment:** The effects of impairment on DSIP exposure and clearance remain insufficiently characterized. [23]

Reported and Possible Side Effects

Small human IV insomnia studies generally described DSIP as well tolerated during brief exposure. These observations concern the monitored populations and routes. [4–7,13,23]

- **Headache, nausea, vertigo, perspiration, and hypotension:** In an older IV withdrawal-treatment series of 107 patients, nine experienced minor transient effects and three had more serious reactions, including two cases of hypotension. One patient developed

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progressive hypotension after a second injection. The withdrawal setting and limited reporting complicate causal attribution and do not establish event rates for SC or intranasal use. [23]

- **Autonomic changes during anesthesia:** In a randomized study of 24 women, IV DSIP increased heart rate and reduced heart-rate variability during isoflurane anesthesia, alongside EEG and bispectral-index findings consistent with lighter anesthesia. [21]

Diminished Response and Cycling

A specific DSIP receptor has not been identified. Waning responsiveness during repeated SC use remains speculative. [19] Physio

Use with Low Dose Epitalon

The speculative, experimental low-dose Epitalon schedule discussed for sleep is

• 50–300 mcg intranasally before bed

Epitalon has a plausible mechanistic connection to circadian regulation: experiments reported altered pineal markers in stressed rats and increased evening melatonin in old monkeys. [17,22,29] However, it did not significantly alter melatonin secretion in isolated pineal glands from young or old rats, indicating context-dependent effects. [28] Sleep benefits from low-dose nightly administration remain speculative.

Combining low-dose DSIP with low-dose Epitalon is a speculative experimental practice.

Biological Mechanism

DSIP-like immunoreactivity has been detected in mammalian tissues and fluids, but the identity, synthesis, and physiological role of an endogenous DSIP molecule remain unresolved. A specific receptor has not been identified. [9,19] Experimental studies report effects on sleep-related electrophysiology, neuroendocrine signaling, cellular stress responses, and autonomic activity. [12,16,18,20,21] These findings support several plausible, context-dependent mechanisms rather than a single validated pathway.

Neural signaling: In rat neurons, DSIP potentiated GABA-activated currents and modified N-methyl-D-aspartate (NMDA)/glutamate signaling. [20] Rat EEG studies reported increased delta-frequency activity. [16] Modulation of inhibitory–excitatory balance is therefore a plausible mechanism for sleep-related effects in humans.

Stress-axis signaling: A small randomized human crossover study found decreased ACTH-like immunoreactivity with unchanged cortisol. [12] This supports a specific neuroendocrine effect, rather than a demonstrated reduction in adrenal cortisol output. Broader HPA modulation remains a plausible mechanism for stress-related effects.

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Cellular stress and neuroprotection: In specific rat models, DSIP reduced neurological impairment and mortality after cerebral ischemia and preserved brain-mitochondrial oxidative phosphorylation during hypoxia. [14,18] Rat studies also reported altered glutamate signaling [20] and lipid-peroxidation markers under hyperoxic conditions. [15] A later intranasal rat stroke study found motor improvement over time within the DSIP group, but no significant reduction in infarct volume or rotation asymmetry versus controls. [24] Neuroprotection is therefore supported in particular preclinical settings; broader human recovery and resilience benefits remain plausible. [31]

Autonomic signaling: IV DSIP increased heart rate and reduced heart-rate variability during isoflurane anesthesia, findings consistent with reduced parasympathetic activity in that setting. [21] Effects on autonomic regulation during ordinary sleep are plausible.

References

[1] Schoenenberger GA, Monnier M. Characterization of a delta-electroencephalogram (-sleep)-inducing peptide. *Proc Natl Acad Sci U S A*. 1977;74(3):1282-1286. doi:10.1073/pnas.74.3.1282

[2] Schoenenberger GA, Maier PF, Tobler HJ, Wilson K, Monnier M. The delta EEG (sleep)-inducing peptide (DSIP). XI. Amino-acid analysis, sequence, synthesis and activity of the nonapeptide. *Pflugers Arch*. 1978;376(2):119-129. doi:10.1007/bf00581575

[3] Schneider-Helmert D, Graf M, Schoenenberger GA. Synthetic delta-sleep-inducing peptide improves sleep in insomniacs. *Lancet*. 1981;1(8232):1256-1257. doi:10.1016/s0140-6736(81)92417-x

[4] Schneider-Helmert D, Gnirss F, Monnier M, Schenker J, Schoenenberger GA. Acute and delayed effects of DSIP (delta sleep-inducing peptide) on human sleep behavior. *Int J Clin Pharmacol Ther Toxicol*. 1981;19(8):341-345. PubMed PMID:6895513

[5] Schneider-Helmert D, Schoenenberger GA. The influence of synthetic DSIP (delta-sleep-inducing-peptide) on disturbed human sleep. *Experientia*. 1981;37(9):913-917. doi:10.1007/bf01971753

[6] Schneider-Helmert D, Schoenenberger GA. Effects of DSIP in man. Multifunctional psychophysiological properties besides induction of natural sleep. *Neuropsychobiology*. 1983;9(4):197-206. doi:10.1159/000117964

[7] Kaeser HE. A clinical trial with DSIP. *Eur Neurol*. 1984;23(5):386-388. doi:10.1159/000115717

[8] Graf MV, Kastin AJ. Delta-sleep-inducing peptide (DSIP): a review. *Neurosci Biobehav Rev*. 1984;8(1):83-93. doi:10.1016/0149-7634(84)90022-8

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[9] Schoenenberger GA. Characterization, properties and multivariate functions of delta-sleep-inducing peptide (DSIP). *Eur Neurol.* 1984;23(5):321-345. doi:10.1159/000115711

[10] Schneider-Helmert D. Effects of delta-sleep-inducing peptide on 24-hour sleep-wake behaviour in severe chronic insomnia. *Eur Neurol.* 1987;27(2):120-129. doi:10.1159/000116143

[11] Graf MV, Saegesser B, Schoenenberger GA. Degradation and aggregation of delta sleep-inducing peptide (DSIP) and two analogs in plasma and serum. *Peptides.* 1987;8(4):599-603. doi:10.1016/0196-9781(87)90031-3

[12] Bjartell A, Ekman R, Bergquist S, Widerlöv E. Reduction of immunoreactive ACTH in plasma following intravenous injection of delta sleep-inducing peptide in man. *Psychoneuroendocrinology.* 1989;14(5):347-355. doi:10.1016/0306-4530(89)90004-8

[13] Bes F, Hofman W, Schuur J, Van Boxtel C. Effects of delta sleep-inducing peptide on sleep of chronic insomniac patients. A double-blind study. *Neuropsychobiology.* 1992;26(4):193-197. doi:10.1159/000118919

[14] Shandra AA, Godlevskii LS, Brusentsov AI, et al. Effects of delta-sleep-inducing peptide in cerebral ischemia in rats. *Neurosci Behav Physiol.* 1998;28(4):443-446. doi:10.1007/bf02464804

[15] Lysenko AV, Alperovich DV, Uskova NI, Mendzheritsky AM. Metabolic features of the adaptive effect of delta-sleep inducing peptide and piracetam under hyperoxic conditions. *Biochemistry (Mosc).* 1999;64(6):652-657. PubMed PMID:10395980

[16] Stanojlović OP, Zivanović DP, Susić VT. The effect of delta sleep-inducing peptide on the EEG and power spectra in rat. *Indian J Physiol Pharmacol.* 2000;44(4):428-434. PubMed PMID:11214497

[17] Sibarov DA, Kovalenko RI, Malinin VV, Khavinson VKh. Epitalon influences pineal secretion in stress-exposed rats in the daytime. *Neuro Endocrinol Lett.* 2002;23(5-6):452-454. PubMed PMID:12500171

[18] Khvatova EM, Samartzev VN, Zagoskin PP, Prudchenko IA, Mikhaleva II. Delta sleep inducing peptide (DSIP): effect on respiration activity in rat brain mitochondria and stress protective potency under experimental hypoxia. *Peptides.* 2003;24(2):307-311. doi:10.1016/s0196-9781(03)00040-8

[19] Kovalzon VM, Strekalova TV. Delta sleep-inducing peptide (DSIP): a still unresolved riddle. *J Neurochem.* 2006;97(2):303-309. doi:10.1111/j.1471-4159.2006.03693.x

This document provides educational summaries of limited and evolving peptide data. Nothing here is medical advice for diagnosis or treatment. © 2026 Researcher6076

[20] Grigor'ev VV, Ivanova TA, Kustova EA, Petrova LN, Serkova TP, Bachurin SO. Effects of delta sleep-inducing peptide on pre- and postsynaptic glutamate and postsynaptic GABA receptors in neurons of the cortex, hippocampus, and cerebellum in rats. *Bull Exp Biol Med*. 2006;142(2):186-188. doi:10.1007/s10517-006-0323-9

[21] Pomfrett CJ, Dolling S, Anders NR, Glover DG, Bryan A, Pollard BJ. Delta sleep-inducing peptide alters bispectral index, the electroencephalogram and heart rate variability when used as an adjunct to isoflurane anaesthesia. *Eur J Anaesthesiol.* 2009;26(2):128-134. doi:10.1097/eja.0b013e32831c8644

[22] Araj SK, Brzezik J, Mądra-Gackowska K, Szeleszczuk Ł. Overview of Epitalon-Highly Bioactive Pineal Tetrapeptide with Promising Properties. *Int J Mol Sci.* 2025;26(6):2691. doi:10.3390/ijms26062691

[23] US Food and Drug Administration. Pharmacy Compounding Advisory Committee briefing document: emideltide (free base) and emideltide acetate. July 23–24, 2026. Accessed October 18, 2026. Source

[24] Tukhovskaya EA, Ismailova AM, Shaykhutdinova ER, et al. Delta Sleep-Inducing Peptide Recovers Motor Function in SD Rats after Focal Stroke. *Molecules.* 2021;26(17):5173. doi:10.3390/molecules26175173

[25] Popovich IG, Voitenkov BO, Anisimov VN, et al. Effect of delta-sleep inducing peptide-containing preparation Deltaran on biomarkers of aging, life span and spontaneous tumor incidence in female SHR mice. *Mech Ageing Dev.* 2003;124(6):721-731. doi:10.1016/s0047-6374(03)00082-4 Open full text

[26] Eunice Kennedy Shriver National Institute of Child Health and Human Development. What happens during sleep? Reviewed April 29, 2019. Accessed October 18, 2026. Source

[27] Van Cauter E, Leproult R, Plat L. Age-related changes in slow wave sleep and REM sleep and relationship with growth hormone and cortisol levels in healthy men. *JAMA*. 2000;284(7):861-868. doi:10.1001/jama.284.7.861 PubMed

[28] Djeridane Y, Khavinson VKh, Anisimov VN, Touitou Y. Effect of a synthetic pineal tetrapeptide (Ala-Glu-Asp-Gly) on melatonin secretion by the pineal gland of young and old rats. *J Endocrinol Invest.* 2003;26(3):211-215. doi:10.1007/bf03345159

[29] Goncharova ND, Khavinson BK, Lapin BA. Regulatory effect of Epithalon on production of melatonin and cortisol in old monkeys. *Bull Exp Biol Med.* 2001;131(4):394-396. doi:10.1023/a:1017928925177

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[30] Monti JM, Debellis J, Alterwain P, Pellejero T, Monti D. Study of delta sleep-inducing peptide efficacy in improving sleep on short-term administration to chronic insomniacs. *Int J Clin Pharmacol Res*. 1987;7(2):105-110. PubMed PMID:3583493

[31] Tukhovskaya EA, Shaykhutdinova ER, Ismailova AM, et al. DSIP-Like KND Peptide Reduces Brain Infarction in C57Bl/6 and Reduces Myocardial Infarction in SD Rats When Administered during Reperfusion. *Biomedicines*. 2021;9(4):407. doi:10.3390/biomedicines9040407